

Lecture 6

Building Persuasive Arguments

Why Persuasion is Important

- You need to convince stakeholders to follow your recommendations
 - Data science is only valuable if it has an impact (*i.e.* decisions changes are made a result of your analysis)
 - The people who will make those decisions won't understand your Jupyter notebook, you need to explain it to them
- You need to compete for resources
 - Business stakeholders will usually have a lot of things going on, of which your work is only one
 - You need to convince them that what you've done is valuable (so that they keep paying you)
- You want to establish yourself as an authority
 - To become more senior, you need to be seen as an advisor rather than a technical contributor

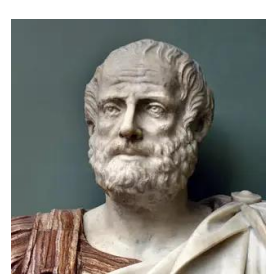
Modes of Persuasion

- Pathos
 - Appeals to the emotions and ideals of the audience
- Ethos
 - The credibility of the speaker
- Logos
 - Logical appeal, usually referring to facts and figures of the speaker
- A persuasive argument will use all three of these modes

Structures of Persuasive Arguments

- Introduction -> Background -> Proposition -> Proof (Classical Method)

- Introduction: Introduce the issue and its significance
- Background: Information that provides better understanding of the issue
- Proposition: What to do about the problem based on this evidence
- Proof: Explanation of your reasons and evidence



- Claim -> Evidence -> Warrant (Toulmin Method)

- Claim: The main idea or recommendation.
- Evidence: Supporting data and examples.
- Warrant: Explanation of why the evidence supports the claim.



Classical Structure

My wife and I are debating what movie to watch

- Problem: We need to find a movie to watch and there's a lot of options available to stream. We'd like to watch a comedy and something familiar
- Background:
 - There's six movies we've watched recently that have sequels
 - Of these, three are comedies
 - Of these three, Paddington 2 has the best reviews
- Proposition: We should watch Paddington 2.
- Proof: It fits what we want to watch and it has great reviews.

Toulmin Structure

My wife and I are debating what movie to watch

- Claim: I think tonight we should watch Paddington 2
- Evidence:
 - Paddington 2 has a 99% fresh rating on Rotten Tomatoes
 - We watched Paddington 1 and enjoyed it
 - Finley likes his stuffed Paddington
- Warrant: This movie has good reviews, is a sequel to something we enjoyed, and has a distinct link to our family.

When to Use Each Structure

Feature	Toulmin Argument	Classical Argument
Best For	• Complex, nuanced issues requiring deep analysis.	• Persuading a broad audience with a clear, direct argument.
Strengths	• Transparency of reasoning • Introducing the claim early sets context for the evidence	• Storytelling approach is easier to follow • Can allow listeners to independently reach the conclusion
Primary Weakness	• Potential for audience confusion • Introducing the claim early can distract from the evidence	• Audience may lose interest before hearing the recommendation

Exercise – Answering an Interview Question

A very common question at job interviews is:

“Why should I hire you”

“What makes you a good fit for this role”

etc.

Take some time to outline, using either structure, outline the components of your persuasive argument

Presenting Strong Evidence

- Evidence should support your claim
- The audience shouldn't have to work to understand your data
- Each piece of evidence will appeal to *at least* one mode of persuasion, but probably more (even all three!)

Tailoring to Your Audience

- Different audiences will be influenced by different things
 - Managers: a few key performance indicators
 - *e.g.* we can decrease our inventory carrying cost by \$10M per year
 - Quantitative People: precise metrics, comparisons between options
 - *e.g.* we can reduce the inventory turnover ratio by between 6% and 23% compared to the current replenishment rule
 - Laypeople: Easy-to-understand comparison, little/no specifics
 - *e.g.* we reduce the length of time items sit on store shelves
- If you have multiple audiences, you will need to be ready to address them all

Strong Evidence – Example 1

Models		PatchTST/64		PatchTST/42		DLinear		FEDformer		Autoformer		Informer		Pyraformer		LogTrans	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Weather	96	0.149	0.198	<u>0.152</u>	<u>0.199</u>	0.176	0.237	0.238	0.314	0.249	0.329	0.354	0.405	0.896	0.556	0.458	0.490
	192	0.194	0.241	<u>0.197</u>	<u>0.243</u>	0.220	0.282	0.275	0.329	0.325	0.370	0.419	0.434	0.622	0.624	0.658	0.589
	336	0.245	0.282	<u>0.249</u>	<u>0.283</u>	0.265	0.319	0.339	0.377	0.351	0.391	0.583	0.543	0.739	0.753	0.797	0.652
	720	0.314	0.334	<u>0.320</u>	<u>0.335</u>	0.323	0.362	0.389	0.409	0.415	0.426	0.916	0.705	1.004	0.934	0.869	0.675
Traffic	96	0.360	0.249	<u>0.367</u>	<u>0.251</u>	0.410	0.282	0.576	0.359	0.597	0.371	0.733	0.410	2.085	0.468	0.684	0.384
	192	0.379	0.256	<u>0.385</u>	<u>0.259</u>	0.423	0.287	0.610	0.380	0.607	0.382	0.777	0.435	0.867	0.467	0.685	0.390
	336	0.392	0.264	<u>0.398</u>	<u>0.265</u>	0.436	0.296	0.608	0.375	0.623	0.387	0.776	0.434	0.869	0.469	0.734	0.408
	720	0.432	0.286	<u>0.434</u>	<u>0.287</u>	0.466	0.315	0.621	0.375	0.639	0.395	0.827	0.466	0.881	0.473	0.717	0.396
Electricity	96	0.129	0.222	<u>0.130</u>	<u>0.222</u>	0.140	0.237	0.186	0.302	0.196	0.313	0.304	0.393	0.386	0.449	0.258	0.357
	192	0.147	0.240	<u>0.148</u>	<u>0.240</u>	0.153	0.249	0.197	0.311	0.211	0.324	0.327	0.417	0.386	0.443	0.266	0.368
	336	0.163	0.259	<u>0.167</u>	<u>0.261</u>	0.169	0.267	0.213	0.328	0.214	0.327	0.333	0.422	0.378	0.443	0.280	0.380
	720	0.197	0.290	<u>0.202</u>	<u>0.291</u>	0.203	0.301	0.233	0.344	0.236	0.342	0.351	0.427	0.376	0.445	0.283	0.376
ILI	24	1.319	0.754	<u>1.522</u>	<u>0.814</u>	2.215	1.081	2.624	1.095	2.906	1.182	4.657	1.449	1.420	2.012	4.480	1.444
	36	<u>1.579</u>	<u>0.870</u>	1.430	0.834	1.963	0.963	2.516	1.021	2.585	1.038	4.650	1.463	7.394	2.031	4.799	1.467
	48	1.553	0.815	<u>1.673</u>	<u>0.854</u>	2.130	1.024	2.505	1.041	3.024	1.145	5.004	1.542	7.551	2.057	4.800	1.468
	60	1.470	0.788	<u>1.529</u>	<u>0.862</u>	2.368	1.096	2.742	1.122	2.761	1.114	5.071	1.543	7.662	2.100	5.278	1.560
ETTth1	96	0.370	0.400	<u>0.375</u>	0.399	0.375	0.399	0.376	0.415	0.435	0.446	0.941	0.769	0.664	0.612	0.878	0.740
	192	<u>0.413</u>	0.429	0.414	<u>0.421</u>	0.405	0.416	0.423	0.446	0.456	0.457	1.007	0.786	0.790	0.681	1.037	0.824
	336	0.422	0.440	<u>0.431</u>	0.436	0.439	0.443	0.444	0.462	0.486	0.487	1.038	0.784	0.891	0.738	1.238	0.932
	720	0.447	0.468	<u>0.449</u>	0.466	0.472	0.490	0.469	0.492	0.515	0.517	1.144	0.857	0.963	0.782	1.135	0.852
ETTth2	96	0.274	<u>0.337</u>	0.274	0.336	0.289	0.353	0.332	0.374	0.332	0.368	1.549	0.952	0.645	0.597	2.116	1.197
	192	<u>0.341</u>	<u>0.382</u>	0.339	0.379	0.383	0.418	0.407	0.446	0.426	0.434	3.792	1.542	0.788	0.683	4.315	1.635
	336	0.329	<u>0.384</u>	<u>0.331</u>	0.380	0.448	0.465	0.400	0.447	0.477	0.479	4.215	1.642	0.907	0.747	1.124	1.604
	720	0.379	0.422	0.379	0.422	0.605	0.551	0.412	0.469	0.453	0.490	3.656	1.619	0.963	0.783	3.188	1.540
ETTm1	96	<u>0.293</u>	0.346	0.290	0.342	0.299	<u>0.343</u>	0.326	0.390	0.510	0.492	0.626	0.560	0.543	0.510	0.600	0.546
	192	<u>0.333</u>	0.370	0.332	<u>0.369</u>	0.335	0.365	0.365	0.415	0.514	0.495	0.725	0.619	0.557	0.537	0.837	0.700
	336	<u>0.369</u>	<u>0.392</u>	0.366	<u>0.392</u>	<u>0.369</u>	0.386	0.392	0.425	0.510	0.492	1.005	0.741	0.754	0.655	1.124	0.832
	720	0.416	0.420	<u>0.420</u>	<u>0.424</u>	<u>0.425</u>	<u>0.421</u>	0.446	0.458	0.527	0.493	1.133	0.845	0.908	0.724	1.153	0.820
ETTm2	96	<u>0.166</u>	<u>0.256</u>	0.165	0.255	0.167	0.260	0.180	0.271	0.205	0.293	0.355	0.462	0.435	0.507	0.768	0.642
	192	<u>0.223</u>	<u>0.296</u>	0.220	0.292	0.224	0.303	0.252	0.318	0.278	0.336	0.595	0.586	0.730	0.673	0.989	0.757
	336	0.274	0.329	<u>0.278</u>	0.329	0.281	0.342	0.324	0.364	0.343	0.379	1.270	0.871	1.201	0.845	1.334	0.872
	720	0.362	0.385	<u>0.367</u>	0.385	0.397	0.421	0.410	0.420	0.414	0.419	3.001	1.267	3.625	1.451	3.048	1.328

Table 3: Multivariate long-term forecasting results with supervised PatchTST. We use prediction lengths $T \in \{24, 36, 48, 60\}$ for ILI dataset and $T \in \{96, 192, 336, 720\}$ for the others. The best results are in **bold** and the second best are underlined.

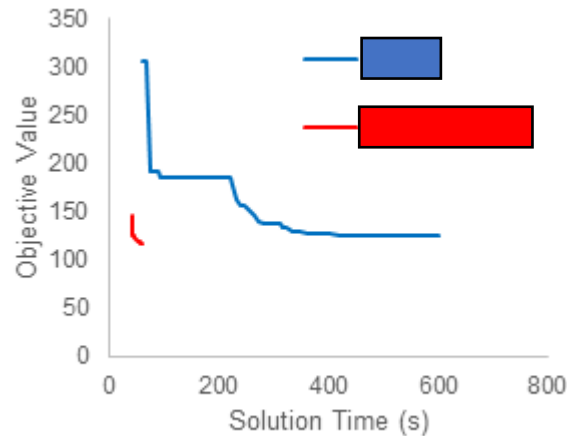
Table from a paper at a large machine learning conference

Who is (are) the audience(s) for this evidence?

How did the author tailor their evidence to this audience?

What modes of persuasion is the author using here?

Strong Evidence – Example 2



Excerpt from a report prepared for a customer interested in the performance of one method (red) against their incumbent method (blue)

Who is (are) the audience(s) for this evidence?

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What modes of persuasion is the author using here?

Strong Evidence – Example 3

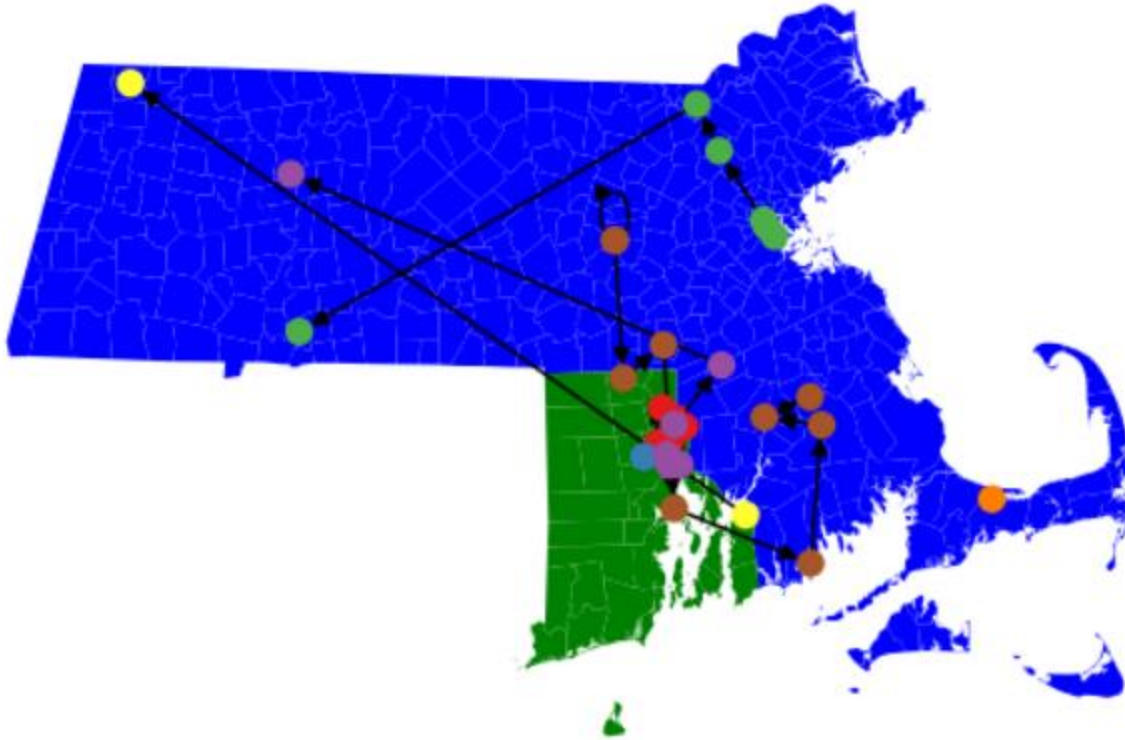


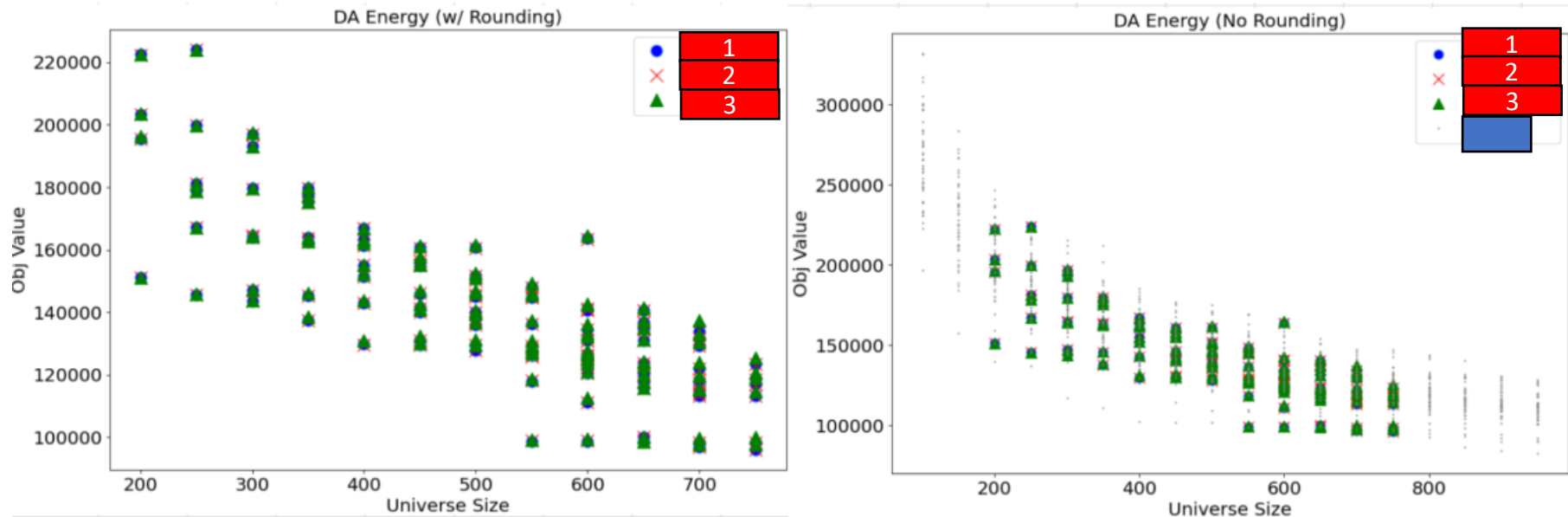
Figure from a presentation on a field service agent scheduling solution

Who is (are) the audience(s) for this evidence?

How did the author tailor their evidence to this audience?

What modes of persuasion is the author using here?

Weak Evidence – Example 1



What make this evidence weak compared to the evidence on previous pages?

How could it be made stronger?

Takeaways

- Being persuasive is important!
 - Ensures your work is impactful
 - Builds your professional credibility
- You can persuade people in three ways
 - Appeals to emotion, such as demonstrations and case studies
 - Appeals to logic, such as statistics and metrics
 - Appeals to credibility, showing the depth and breadth of your knowledge
- There are two very effective structures for a persuasive argument
 - Classical (Introduction -> Background -> Proposition -> Proof)
 - Toulmin (Claim -> Evidence -> Warrant)

Activity

You will find the data on Canvas

- The data provides measurements of 200 test cookies that a bakery produced to try out a new recipe
- Each cookie has three measurements – sugar content, moisture, and density
- The bakery has asked you to recommend which recipe they should put into production

Using this data you will argue one of two claims:

- If your SIS ID is *EVEN* – Recipe A should be put into production
- If your SIS ID is *ODD* – Recipe B should be put into production

Your argument should include **at least one** piece of evidence and a warrant (explanation) for why that evidence supports your claim.